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# Measuring all the noises of LLM Evals

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FAIR at Meta

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# Overview

1. Paired comparisons increase the power of statistical analysis
2. Total noise between A and B is roughly fixed per benchmark, and is governed by  $\text{Var}(A - B) \approx p(1 - p) = \text{Var}(A)$ 
  - (because question-difficulty is  $u_i \sim \text{Beta}(p, 1 - p)$ )
3. Breakdown of total noise to data noise and prediction noise

1. Paired comparisons increase the power of statistical analysis

# Standard Error

- When comparing two LLMs with eval scores A and B, we care about the standard error more than the variance

$$SE(A - B) = \sqrt{\text{Var}(A - B)/N}$$

- Variance measures the spread of A-B every time we ask a question
- Standard error measures our uncertainty about the *mean* of A-B

## Independent SE vs Paired SE (paired over questions $x$ )

Independent:  $\text{Var}(A - B) = \text{Var}(A) + \text{Var}(B)$

Paired:  $\text{Var}_x(A - B) = \text{Var}_x(A) + \text{Var}_x(B) - 2 \text{Cov}_x(A, B)$

And perfect correlation would have  $\text{Cov}_x(A, B) = \text{Var}_x(A) = \text{Var}_x(B)$

Paired SE is better because LLMs are positively correlated across questions

(Note there are two sources of randomness: question/data noise based on  $x$ , and prediction noise based on seed  $\epsilon$ , so we actually care about  $\text{Var}_{x,\epsilon}[A(x,\epsilon)] \dots$ )

# LLMs are positively correlated

Good models answer the ‘easy’ questions as well as the ‘hard’ questions

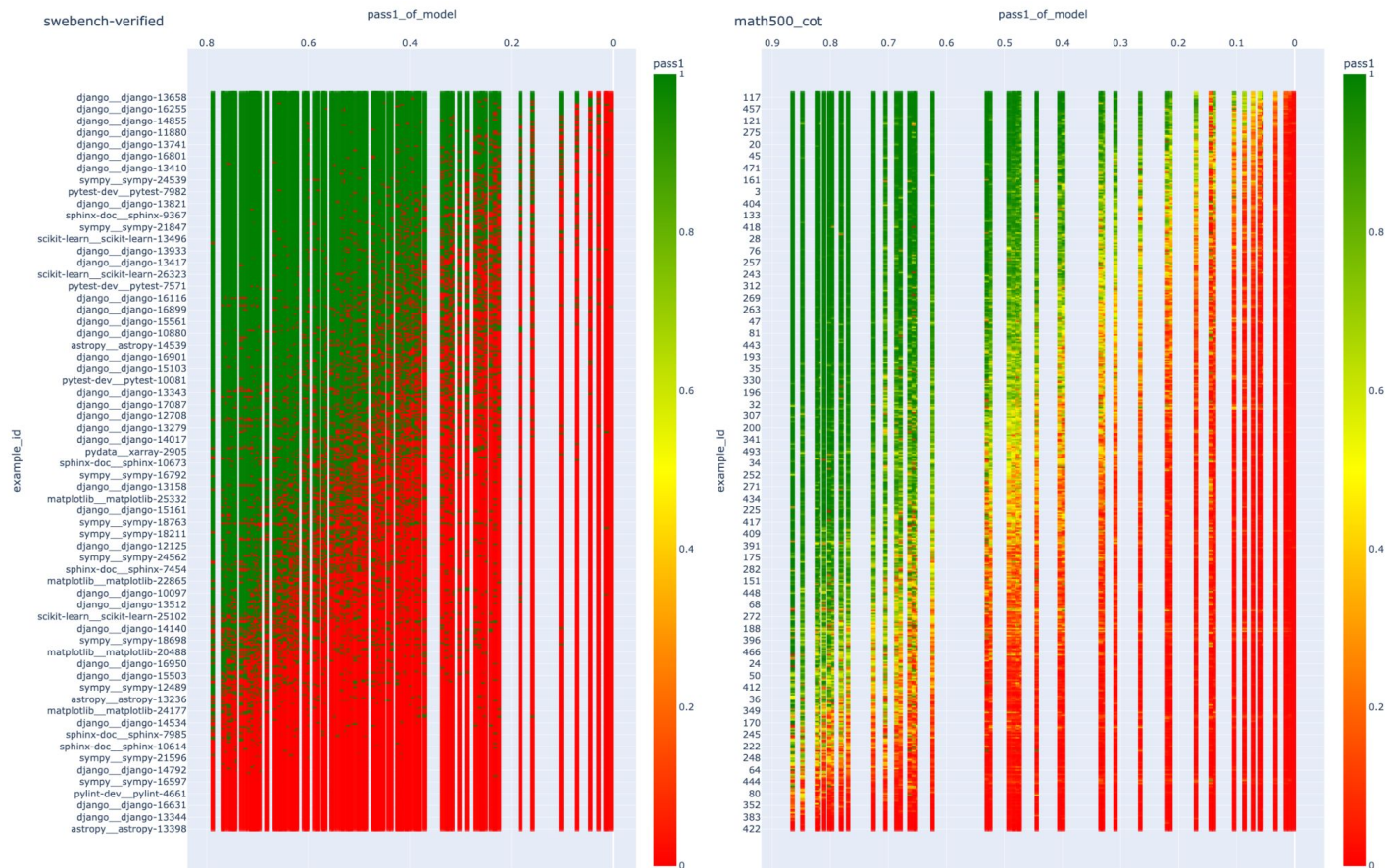


Figure 4: Results heatmap. Each row is a different question, sorted from the easiest to the hardest. Each column is a model whose  $x$ -coordinate is the overall accuracy. Left: SWEbench-verified, 1 prediction per question; Right: MATH500, 1000 predictions per question. See §4.1 for more details.

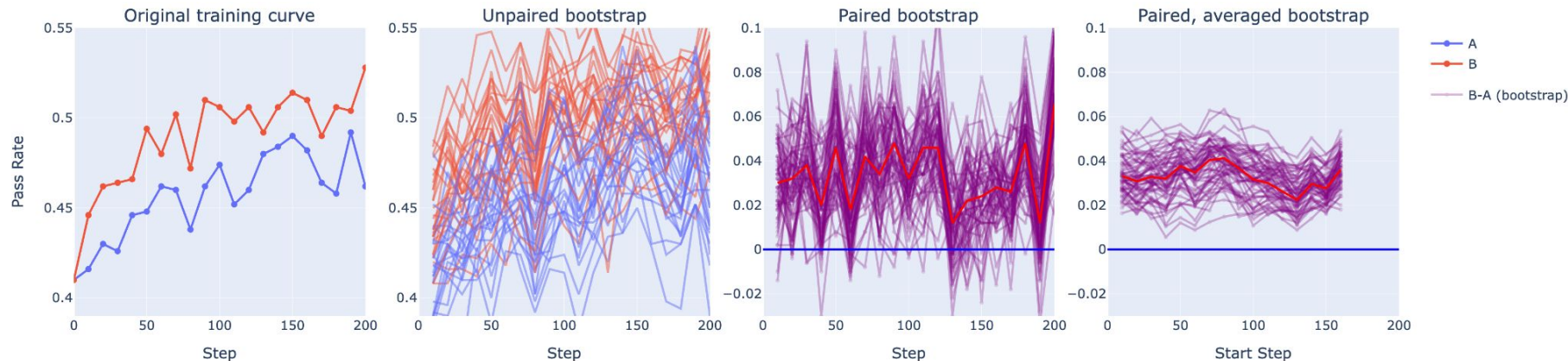
## Example 1 - HumanEval $N = 164$ $\bar{A} = 0.5$

- $SE(A) = \sqrt{1/164 \times 0.5(1 - 0.5)} = 4\%$
- Unpaired  $SE(A-B) = SE(A) \times \sqrt{2} = 6\%$
- Need difference of  $6\% \times 1.96 = 12\%$  to get 0.05 p-value
  
- Paired  $SE(A-B) \sim SE(A) = 4\%$
- Need difference of  $4\% \times 1.96 = 8\%$  to get 0.05 p-value

| Test Type                | Standard error of $\bar{A} - \bar{B}$ | Difference for $p < 0.05$ |
|--------------------------|---------------------------------------|---------------------------|
| Unpaired                 | 6%                                    | 12%                       |
| Unpaired with averaging  | 4%                                    | 8%                        |
| Paired single prediction | 4%                                    | 8%                        |
| Paired with averaging    | 1-2%                                  | 2-4%                      |

“with averaging” means sampling each question  $K \rightarrow \infty$  times

## Example 2 - Training curves on SWEBench-Verified (N=500)



Bootstrap questions to show noise coming from data  $x$  as well as prediction noise  $\epsilon$

- Original curve
  - Suggests  $B > A$
- Unpaired bootstrap (resampling questions for each checkpoint)
  - hard to interpret
- Paired bootstrap (same resampled questions for A,B each checkpoint)
  - $B > A$  has z-score of 1.7 (slightly significant)
- Paired bootstrap, averaging over last 5 checkpoints
  - $B > A$  has z-score of 3.5 (very significant)



## Aside: Pooled z-scores for meta-analysis

$$\text{z-score } z = \frac{\bar{A} - \bar{B}}{\text{SE}[A - B]}$$

- We can combine z-scores across multiple benchmarks

$$\bar{z} = \frac{\sum_i w_i z_i}{(\sum_i w_i^2)^{1/2}}$$

- $w_i = 1$ 
  - Each benchmark has equal weight
- $w_i = N_i^{1/2}$ 
  - Each question has equal weight
  - Most statistical power for a pool of iid questions, but less valid for us since different evals capture different abilities

2. Total noise of  $A - B$  is roughly fixed per benchmark

(and is governed by  $\text{Var}(A - B) \approx p(1 - p) = \text{Var}(A)$ )

Q: Where does this  $p(1-p)$  come from? A: 'Beta Theoretical Model'

- If an LLM has avg acc  $p$ , the expected accuracy on question  $i$  is given by

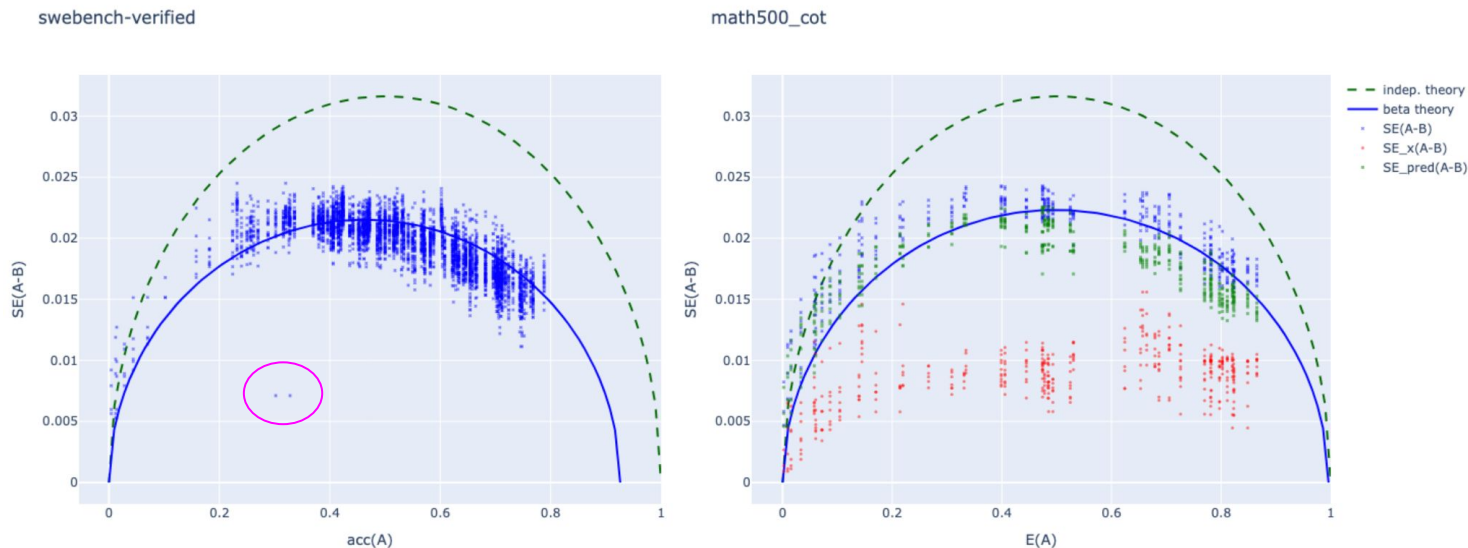
$$u_i \sim \text{Beta}(p, 1 - p)$$

- This gives  $\mathbb{E}[u_i] = p$   $\text{Var}(u_i) = p(1 - p)/2$
- Because we're comparing two models on a single question  $i$ , we have

$$\begin{aligned}\text{Var}_\epsilon(A_i - B_i) &= \text{Var}_\epsilon(A_i) + \text{Var}_\epsilon(B_i) - 2 \text{Cov}_\epsilon(A_i, B_i) \\ &= 2 \text{Var}(u_i) - 0 \\ &= p(1 - p)\end{aligned}$$

- $\text{Cov}_\epsilon(A_i, B_i) = 0$  because LLMs draw independent predictions on a given question  $i$

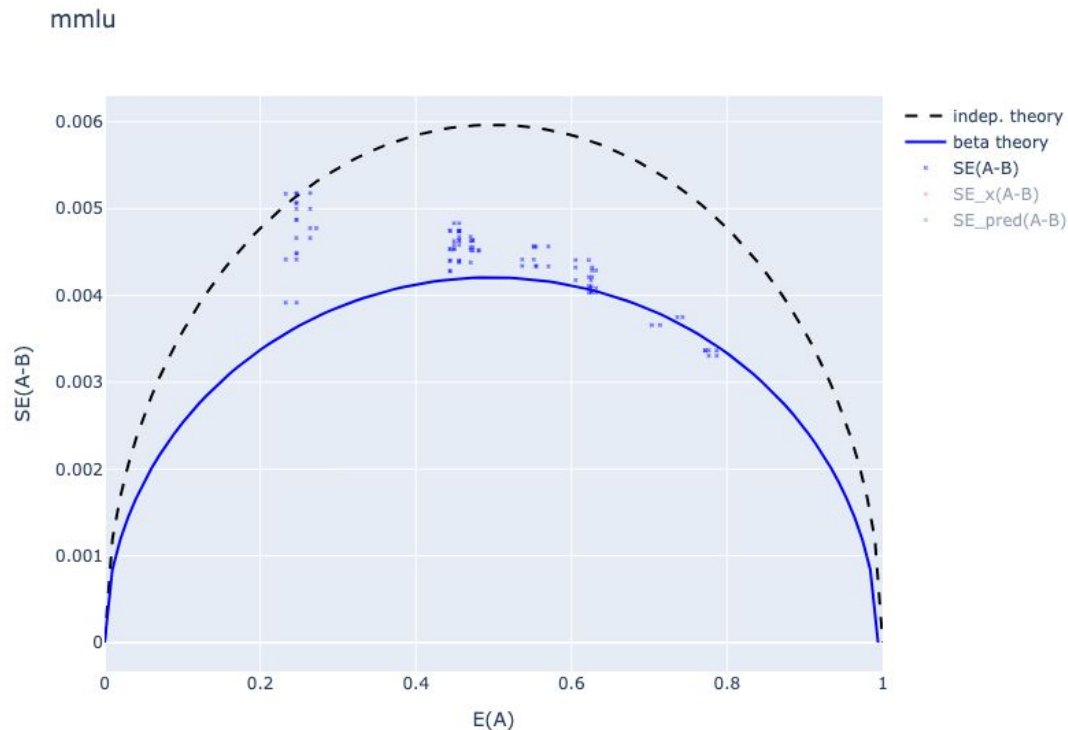
# All-Pairs Analysis (plot all $SE(A-B)$ pairs against $acc(A)$ )



- $Indep = \sqrt{2} \cdot \text{beta-theory}$
- Note: they squash beta-theory to be in  $[0, \text{max\_acc}]$
- If we have a benchmark's SE-curve, we can quickly put together a hypothesis test to check the significance of our results (or others')
- We do see outliers, but often we expect them: e.g. 2 SWE-fixer models

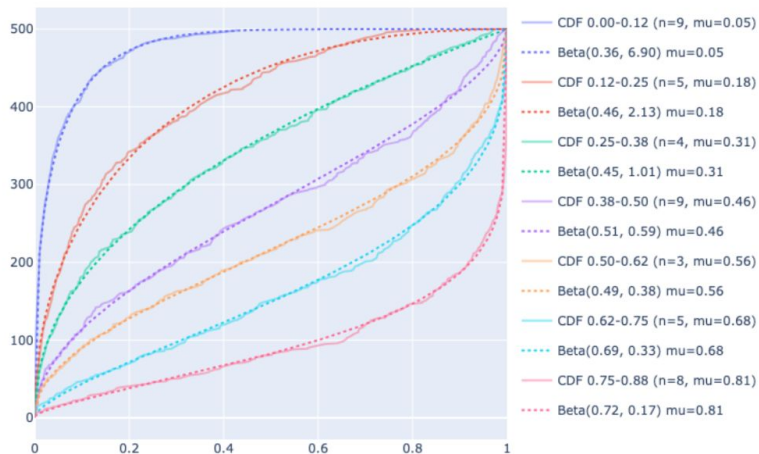
# MMLU Outliers

- Indep theory works better when guessing ( $p=1/4$ ) because luck-noise overrides difficulty-noise

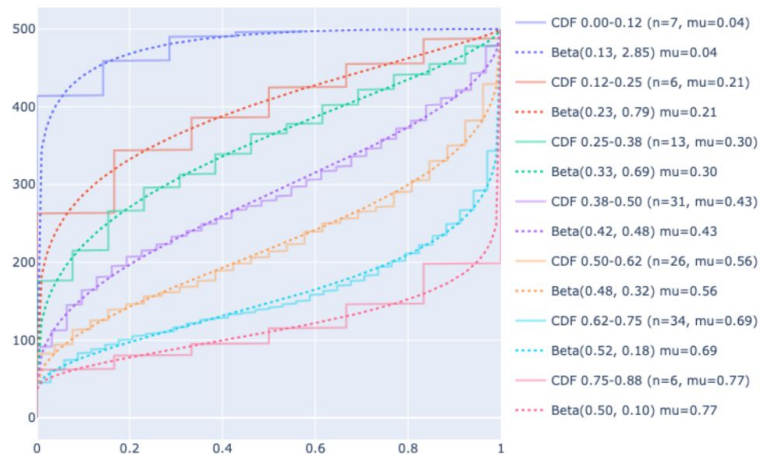


# Evidence for the Beta theory

cdf on math500\_cot



cdf on swebench-verified



- Each line is a group of LLMs binned by similar accuracy
- Cumulative distribution of question-accuracy (solid lines) is matched well by the fitted Beta(a,b) CDFs
  - Although often  $a + b \neq 1$  (but that might be okay – Beta(p,1-p) is the theoretical value)
  - Observation: Larger p  $\rightarrow$  smaller a+b  $\rightarrow$  stronger bimodality (at 0 and 1)  $\rightarrow$  less guessing?

3. Total noise = data noise + prediction noise

$$\text{Var}_{x,\epsilon}[A(x, \epsilon)] = \text{Var}_x[\mathbb{E}_\epsilon[A]] + \mathbb{E}_x[\text{Var}_\epsilon[A]],$$

Total variance = data variance + prediction variance.

What we actually  
measure in experiments

#### Data variance

- Measures how question difficulty varies
- Caused by sampling (N) questions from some 'platonic ideal' pool of questions
- Cannot be fully reduced (but more N\_might\_ improve estimate)

#### Prediction variance

- Measures how a model's answers vary for a given question
- Caused by model stochasticity
- Can be reduced, increase K, change temp



# How to calculate (paired) data noise and prediction noise

Draw K samples for each of N questions

- Total var =  $\text{Var}_{x,\epsilon}[A - B] = \text{Var}_{x,\epsilon}[A] + \text{Var}_{x,\epsilon}[B] - 2\text{Cov}_{x,\epsilon}[A, B]$   
 $= \text{Var}_{x,\epsilon}[A] + \text{Var}_{x,\epsilon}[B] - 2\text{Cov}_x[\text{E}_\epsilon[A], \text{E}_\epsilon[B]]$

- Second term below is 0

- $\text{Cov}_{x,\epsilon}[A, B] = \text{Cov}_x[\text{E}_\epsilon[A], \text{E}_\epsilon[B]] + \text{E}_x[\text{Cov}_\epsilon[A, B]]$

Paired LLM correlation  
means  $\text{Cov}_x \approx \text{Var}_x$   
(so pairing is a good idea)

- Pred var =  $\text{E}_x[\text{Var}_\epsilon[A(x, \epsilon) - B(x, \epsilon')]] = \text{E}_x[\text{Var}_\epsilon[A(x, \epsilon)]] + \text{E}_x[\text{Var}_\epsilon[B(x, \epsilon)]]$

- Data var = total var - pred var

Our beta-theory says  
 $\text{Var}_\epsilon[A(x, \epsilon)] = \text{Var}(u_i) = p(1 - p)/2$

# Small-K Correction: Use the right estimator!

1. We want an estimator for  $\bar{A}_K$  not  $A$ , so divide prediction variance by K

$$\text{Var}_{x,\epsilon}[\bar{A}_K] = \text{Var}_x[\text{E}_\epsilon[\bar{A}_K]] + \text{E}_x[\text{Var}_\epsilon[\bar{A}_K]] = \text{Var}_x[\text{E}_\epsilon[A]] + \frac{1}{K} \text{E}_x[\text{Var}_\epsilon[A]]$$

Total var

Data var

Pred var

2. Actually, divide by K-1, not K (because 1. is actually biased)

$$\text{Var}_x[\text{E}_\epsilon[A]] = \text{Var}_x[\bar{A}_K] - \frac{1}{K} \text{E}_x[\text{Var}_\epsilon[A]] \approx \text{Var}_x[\bar{A}_K] - \frac{1}{K-1} \hat{\text{E}}_x \hat{\text{Var}}_\epsilon[\bar{A}_K]$$

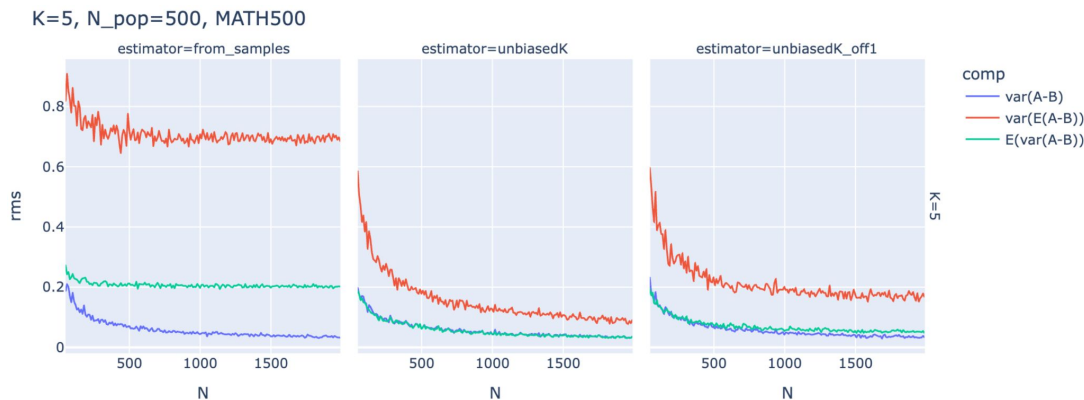
Data var

Total var

Pred var

Simple sample-based estimators

RMS of bootstrapped K=5  
estimate against ground-truth  
(K=1000, N=500)



| Name                | Formula                              | Code                                | Bernoulli                              |
|---------------------|--------------------------------------|-------------------------------------|----------------------------------------|
| total variance      | $\text{Var}_{x,\epsilon}[A]$         | <code>var(A)</code>                 | $\bar{p}(1 - \bar{p})$                 |
| data variance       | $\text{Var}_x[\text{E}_\epsilon[A]]$ | <code>var(mean(A, axis=1))-b</code> | $\frac{1}{N} \sum_i (p_i - \bar{p})^2$ |
| prediction variance | $\text{E}_x[\text{Var}_\epsilon[A]]$ | <code>mean(var(A, axis=1))+b</code> | $\frac{1}{N} \sum_i p_i(1 - p_i)$      |

where  $b=1/(K-1)*\text{mean}(\text{var}(A, \text{axis}=1))$

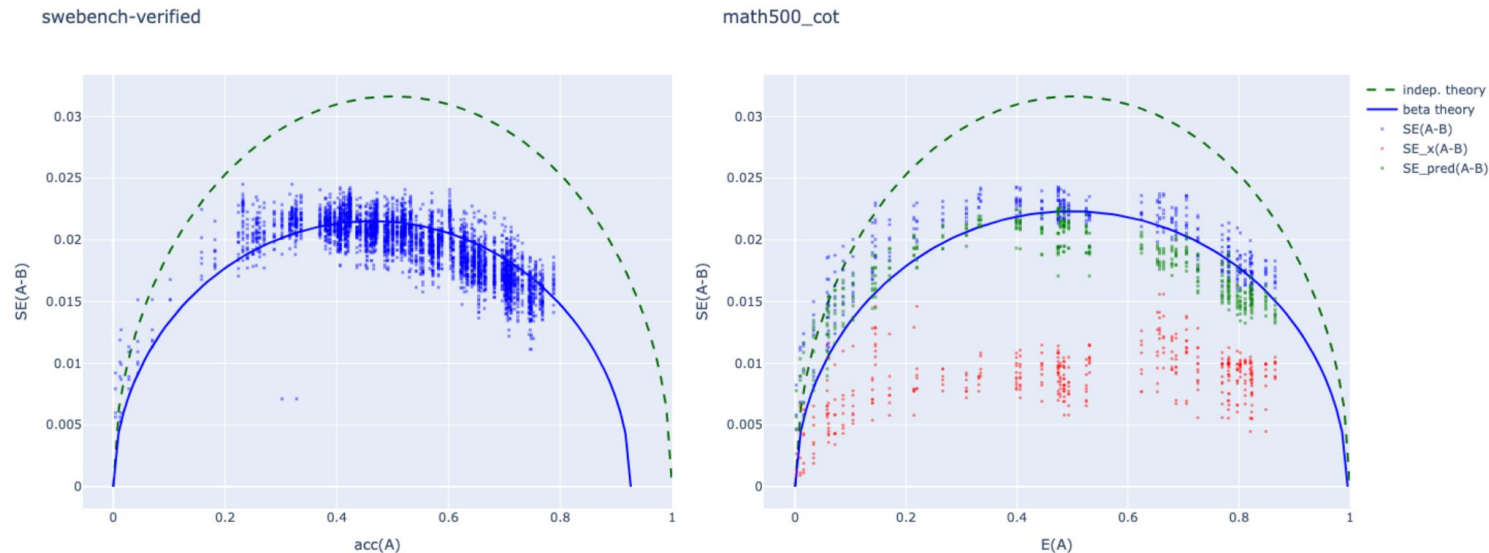
Table 1: Unpaired estimators where  $A \in \mathbb{R}^{N \times K}$  with  $K$  samples for each of  $N$  questions.

| Formula                                  | Code                                                            |
|------------------------------------------|-----------------------------------------------------------------|
| $\text{Var}_{x,\epsilon}[A - B]$         | <code>var(A)+var(B)-2*cov(mean(A,axis=1),mean(B,axis=1))</code> |
| $\text{Var}_x[\text{E}_\epsilon[A - B]]$ | <code>var(mean(A,axis=1)-mean(B,axis=1))-b</code>               |
| $\text{E}_x[\text{Var}_\epsilon[A - B]]$ | <code>mean(var(A,axis=1)+var(B,axis=1)) +b</code>               |

where  $b=1/(k_A-1)*\text{mean}(\text{var}(A, \text{axis}=1))+1/(k_B-1)*\text{mean}(\text{var}(B, \text{axis}=1))$

Table 2: Paired estimators where  $A \in \mathbb{R}^{N \times K_A}$ ,  $B \in \mathbb{R}^{N \times K_B}$ .

# Observation: Usually, data noise < pred noise

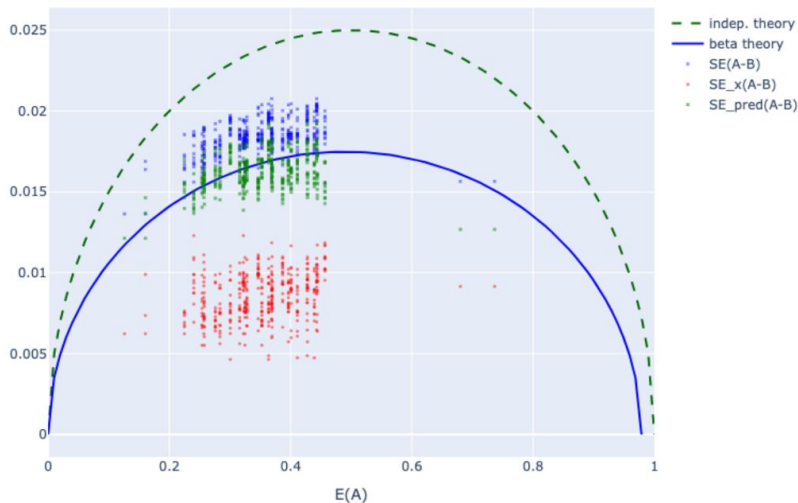


We can look at our noise decomposition and see whether it'd be better for us to:

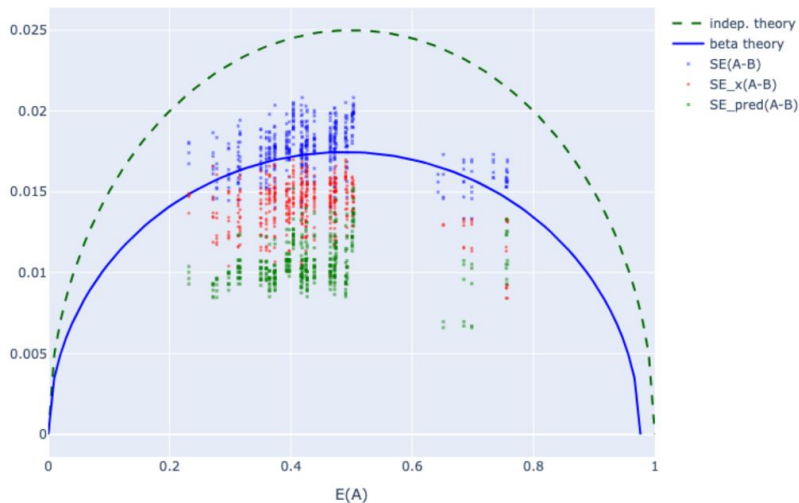
- Decrease data noise (by increasing  $N$ )
- Prediction noise (by increasing  $K$ )

# Changing temperature

CRUXEval-input-T0.8



CRUXEval-input-T0.2



Temperature controls a tradeoff between the two noises:

- Lower temperature reduces prediction noise but increases data noise because model commits to most likely answer
- ...but don't expect the total noise to change much (it's 'fully determined' by  $p$ )

# Conclusion

# Conclusion

1. Use paired analysis (looking at A and B's per-question accuracies)
2. Do quick hypothesis testing of  $A > B$  with  $SE(A - B) \approx SE(A) \approx \sqrt{p(1-p)/N}$ 
  - a. Just need to know the benchmark's N, k (and maybe whether to squash the  $p(1-p)$  into  $[0, \text{max\_acc}]$ )
3. Useful conceptual breakdown of total variance
$$\text{Var}_{x, \epsilon}[A(x, \epsilon)] = \text{Var}_x[\mathbb{E}_{\epsilon}[A]] + \mathbb{E}_x[\text{Var}_{\epsilon}[A]],$$

Total variance = data variance + prediction variance.

# Measuring all the noises of LLM evals

[Home](#) [Paper](#) [Code](#)

**Datasets:** [main](#), [train\\_curve](#), Open models with vLLM: [T=0.7](#), [T=1](#), [high K T=0.7](#),

Details are linked and more summary stats are available on mouse over.

**size:** number of questions in the eval

**models:** the number of models used on this eval

**SE(A):** the standard error (SE\_x for data SE) in percent. A difference of  $1.96 \times 1.41 \times \text{SE(A)}$  is the conventional standard

**SE(A-B):** the paired standard error (SE\_x for data SE) in percent. A difference of  $1.96 \times \text{SE(A-B)}$  is the conventional standard

**corr(A,B):** correlation of expected scores in percent

**no\_solve:** percent questions not solved by any models

**tau-:** percent questions negatively correlated with the overall model quality as measured by Kendall's tau

**details:** aggregating by the models; by examples; the prediction heatmap; and raw data tables.

**Raw data:** [summary.csv](#)

| benchmark_id       | size | models | SE(A)       | SE_x(A)     | SE(A-B)     | SE_x(A-B)   | corr(A,B) | no_solve | tau- | details                                                                                        |
|--------------------|------|--------|-------------|-------------|-------------|-------------|-----------|----------|------|------------------------------------------------------------------------------------------------|
| aime2024_cot       | 30   | 51     | <u>7.4</u>  | <u>5.5</u>  | <u>7.6</u>  | <u>3</u>    | <u>65</u> | 37       | 0    | <a href="#">models</a>   <a href="#">examples</a>   <a href="#">data</a>   <a href="#">raw</a> |
| aime2025_cot       | 30   | 52     | <u>7</u>    | <u>6</u>    | <u>6.7</u>  | <u>4.2</u>  | <u>63</u> | 50       | 0    | <a href="#">models</a>   <a href="#">examples</a>   <a href="#">data</a>   <a href="#">raw</a> |
| ap_cot             | 711  | 52     | <u>1.6</u>  | <u>1.1</u>  | <u>1.9</u>  | <u>1.1</u>  | <u>43</u> | 0        | 3.2  | <a href="#">models</a>   <a href="#">examples</a>   <a href="#">data</a>   <a href="#">raw</a> |
| bbh_cot            | 6511 | 45     | <u>0.55</u> | <u>0.37</u> | <u>0.68</u> | <u>0.36</u> | <u>39</u> | 2        | 1.6  | <a href="#">models</a>   <a href="#">examples</a>   <a href="#">data</a>   <a href="#">raw</a> |
| cruxeval_input_cot | 800  | 52     | <u>1.6</u>  | <u>1.1</u>  | <u>1.9</u>  | <u>0.85</u> | <u>49</u> | 2.4      | 1.4  | <a href="#">models</a>   <a href="#">examples</a>   <a href="#">data</a>   <a href="#">raw</a> |



# Reasons to be wary

1. No consideration of clustered questions within a given benchmark
2. Our hypothesis tests here use the central limit theorem which requires large  $N$ 
  - So be careful whenever  $N < \sim 150$  and/or  $p \notin [0.2, 0.8]$
  - Could result in SE that's too large
3. It's unclear (to me) whether the 'beta theory' targets total noise or just prediction noise (which often *does* dominate data noise)
  - Could result in SE that's too small
4. (maybe (i) and (ii) cancel each other out a bit though!)